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EVALUATION OF ADDITIVE CIRCUITS PRINTED WITH SUSTAINABLE AQUEOUS SILVER INKS USING AEROSOL JET PRINTING AND GRAVURE OFFSET PRINTING AND INVESTIGATION OF CIRCUIT REPAIRABILITY

Pradeep Lall
Auburn University
NSF-CAVE3 Electronics Research Center
Department of Mechanical Engineering
Auburn, AL
Email – lall@auburn.edu
Tele: +1(334)844-3424

Ved Soni
Auburn University
NSF-CAVE3 Electronics Research Center
Department of Mechanical Engineering
Auburn, AL

Scott Miller
Nextflex Manufacturing Institute
San Jose, CA

ABSTRACT
Additive printed electronics, in particular, has been a rapidly advancing field, with applications that span from smart packaging to energy harvesting systems to healthcare devices to intelligent textiles. The basic building block of any flexible circuit created by these technologies is conductive ink. This ink typically comprises metal nanoparticles and several solvents. However, many of the solvents used in these inks are toxic substances or include volatile organic compounds (VOCs) that have a variety of negative effects on the environment and human health, including air pollution, smog, respiratory problems, health risks, and reduced air quality. Consequently, there is a strong push toward developing sustainable materials for use in products, including ink formulations with fewer toxic solvents. In this study, researchers investigate the use of two printing techniques, aerosol jet printing and gravure offset printing, to print full-wave rectifiers (AC to DC conversion) and Voltage bridge oscillators (DC to AC conversion) using commercially available aqueous silver ink. The performance of the additively printed circuits is compared to both theoretical values and to the same circuits printed using non-aqueous inks. Additionally, a switch-mode battery charging circuit has also been printed with a sustainable ink and its performance has been investigated.

Keywords: aerosol jet printing, printed electronics, gravure offset printing, sustainability, repairability, aqueous ink, battery charging

NOMENCLATURE

AJP	aerosol jet printing
ECA	electrically conductive adhesive
FWR	full wave rectifier
RMS	root mean square
R/L	resistance per unit length
SM-PMC	switch mode power management circuit
VCO	voltage controlled oscillator
VOC	volatile organic compound

1. INTRODUCTION

The growing need for flexible consumer electronics and the ongoing trend of downsizing and lightweight design are two of the key reasons pushing innovation in additively printed electronics manufacturing [1][2]. However, electronic devices generate a significant portion of the world's e-waste which is alarmingly rising at a pace of 2 Mt annually [3]. Electronic trash (e-waste), which is generated by the increasing use of disposable electronic gadgets, can be decreased by employing printed electronics, which have a huge potential to offer solutions that are biodegradable and recyclable [4]. Additive manufacturing (AM), which uses both conventional and cutting-edge printing techniques, may produce these devices, reducing the quantity of used (and wasted) materials and doing away with the need for etching and masking [5]. Kunnari et al. [6] categorized the advantages of printed electronics as follows: improved recyclability, reduced use of hazardous materials, reduced energy consumption during both the manufacturing and usage phases, and eco-design (efficient use of materials).

The primary elements of printed electronics include substrates consisting of synthetic or natural polymers, as well as inks with conducting, semiconducting, or dielectric

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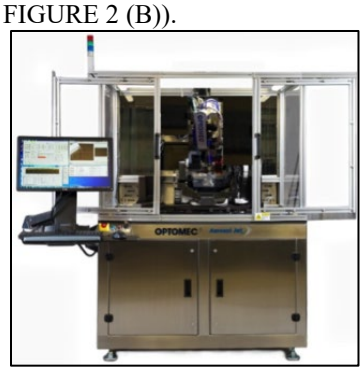
characteristics. Dispersed nanoparticles, dissolved organometallic compounds, and dissolved or dispersed conductive polymers are all forms in which the conducting inks are created [7]. For printing patterns and interconnections, metal nanoparticles have been dispersed in a variety of solvents, including water, isopropyl alcohol, toluene, xylene, and cyclohexanone [8][9][10]. The research trend is shifting toward reducing the use of organic solvents (containing volatile organic compounds (VOCs)) and replacing them with eco-friendly alternatives, based on the environmental and epidemiological issues brought on by VOC emissions which include damage to the ozone layer as well as the human health [11][12]. Therefore, even though water-based inks have more difficult wetting characteristics, it is thought to be more environmentally friendly than organic solvents [13].

This study is focused on examining the usage of sustainable materials, particularly inks for gravure offset and aerosol jet printers. The voltage-controlled oscillator (VCO) and full-wave rectifier (FWR) are the two circuit designs selected for this application. Electrically conductive adhesive (ECA) was used to adhere surface mount parts to the components in these circuits. The capacity to repair products and maintaining their performance almost identical to the performance of the pristine product are additional aspects of sustainable engineering. This was accomplished by manually popping off the surface-mount components fastened to the circuits. Using ECA, new components were reattached to the circuits. Following the repair, the circuit performance was assessed and contrasted with the performance of a brand-new circuit. In order to compare with the sustainable ink, full wave rectifier circuits produced with a non-sustainable ink underwent a similar investigation. Furthermore, a switch mode battery charging circuit has also been designed and printed with a sustainable ink and its performance has been investigated. Section 2 of this paper describes the circuit and the printing methods and procedures, and section 3 details the findings of this study regarding circuits performance and repairability.

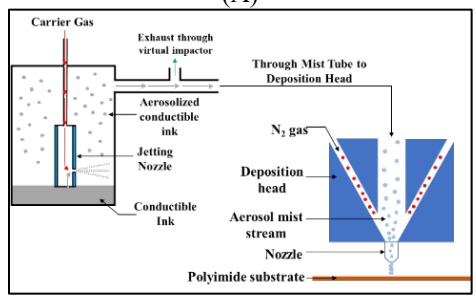
2. MATERIALS AND METHODS

2.1 Additive Printing Equipment

Two printing techniques have been used for this study. Aerosol jet printing which is a thin film deposition method and works with inks having viscosities lesser than 1000 cP, and gravure offset printing which is a thick film deposition method which works with thick pastes. The pneumatic atomizer of the aerosol jet printer has been used for the samples produced in this paper whose process has been highlighted in FIGURE 1 (B). In the pneumatic atomizer, the ink is atomized using a carrier gas pressure which results in an aerosol mist which is then collimated using a sheath gas and then deposited onto the substrate via a nozzle [14]. Gravure offset printing is a modified version of gravure printing which only uses one roller mounted with a blanket used to transfer the pattern from the image plate to the substrate [16][17]. A Pepio F6 Komori gravure offset printer has been used for this study (see FIGURE 2 (A)). This printed uses a pair of blades to stencil the ink onto an image plate. A roller with a mounted blanket rolls on the plate to absorb the image and then re-rolls in the opposite direction on the substrate to transfer the image (see FIGURE 2 (B)).



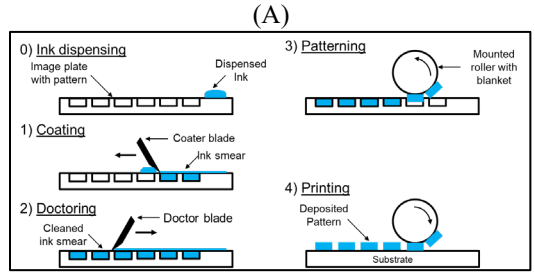
(A)



(B)

FIGURE 1: AEROSOL JET (A) PRINTER AND (B) PRINTING PROCESS WITH PNEUMATIC ATOMIZATION [15]

(A)



(B)

FIGURE 2: GRAVURE OFFSET (A) PRINTER AND (B) PRINTING PROCESS [18]

2.2 Test Vehicle

A circuit that produces a periodic, oscillating signal—often a sine wave, square wave, or triangle wave—is known as an

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electronic oscillator. Direct current (DC) from a power source is converted to alternating current (AC) via oscillators.

TABLE 1: PIN DESCRIPTION OF VOLTAGE-CONTROLLED OSCILLATOR IC

Pin #	Name	Description
1	OE	Output Enable. Drive high to enable the output driver.
2	GND	Ground.
3	SET	Frequency-Setting Input. The voltage on the SET pin (V_{SET}) is regulated to 1V above GND. A resistor connected between SET and GND sets the frequency accurately.
4	DIV	Programmable Divider and Hi-Z Mode Input. A V+ referenced A/D converter monitors the DIV pin voltage (V_{DIV}) to determine a 4-bit result (DIVCODE).
5	V+	Supply Voltage (6V), bypassed directly to the GND pin with a 0.1 uF capacitor
6	OUT	Oscillator Output. The OUT pin swings from GND to V+ with an output resistance of approximately 30Ω

TABLE 2: PIN DESCRIPTION OF SWITCH-MODE PMIC

Pin #	Name	Description
1	COMP	The COMP pin is the control signal of the inner loop of the current mode PWM. Charging begins when the COMP pin reaches 800mV.
2	VCC	Positive Supply Voltage Input
3	GATE	Driver Output for the P-Channel MOSFET
4	GND	Ground
5	CHRG	When the battery is being charged, the CHRG pin is pulled low by an internal N-channel MOSFET. When the charge current drops below the End-of-Charge threshold, the N-channel MOSFET turns off and a 25µA current source is connected from the CHRG pin to GND.
6	BAT	Battery Sense Input. A bypass capacitor of 22µF is required to minimize ripple voltage
7	SNS	Current Amplifier Sense Input. A sense resistor, R_{SNS}, must be connected between the SNS and BAT pins.
8	NTC	NTC (Negative Temperature Coefficient) Thermistor Input. To disable the temperature qualification function, ground the NTC pin.

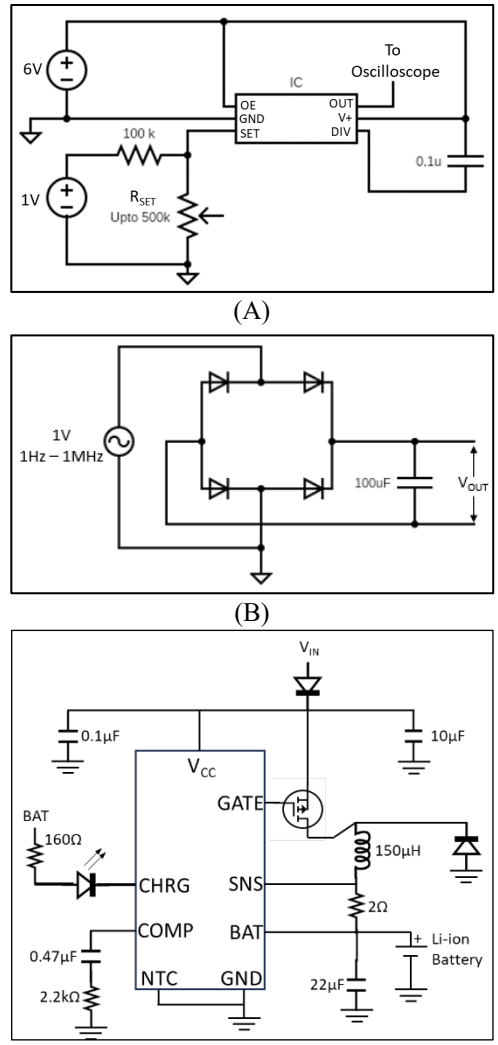

FIGURE 3: (A) VOLTAGE-CONTROLLED OSCILLATOR AND (B) FULL WAVE RECTIFIER CIRCUIT SCHEMATIC

A voltage-controlled oscillator (VCO) is an electronic oscillator whose oscillation frequency is controlled by a voltage input. The applied input voltage determines the frequency of the output oscillation. Frequency modulation (PM) is thus feasible by applying a modulating signal to a VCO's control input. This study designed the circuit using a voltage-controlled oscillator IC that is readily accessible in the market. The selected IC features a 6-pin TSOT-23 architecture (pin description in TABLE 1). FIGURE 3 (A) depicts the circuit diagram created around the IC. The frequency of the output signal from the OUT pin is controlled by the resistance of the potentiometer connected to the SET pin (R_{SET}). The output signal's amplitude is equal to the supply voltage on V+.

The second demonstrator circuit used in this sustainability and repairability investigation is a full wave bridge rectifier. The circuit schematic for the full wave rectifier used in this experiment is shown in FIGURE 3 (B). A sinusoidal signal with 1V amplitude and frequencies ranging from 1 Hz to 100 kHz is utilized as an input, and four diodes are arranged in a bridge

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configuration. A 100 µF capacitor is additionally employed to amplify the rectified signal.

The output signal frequency response to the R_{SET} potentiometer resistance served as the assessment criterion for the voltage-controlled oscillator circuit. In contrast, the RMS value output DC voltage signal for the full wave rectifier circuit was employed as an evaluation metric and contrasted with the theoretical RMS output voltage for various input signal frequencies. The performance of the circuits after repair was another goal of this study's second objective. The components of the pristine circuits were physically removed following their initial evaluation for the repairability inquiry. To finish the 'post-repair' circuit, ECA was once again applied to the circuit pads and fresh parts were attached. The identical tests that were performed on the pristine circuit were then performed on the repaired circuit.

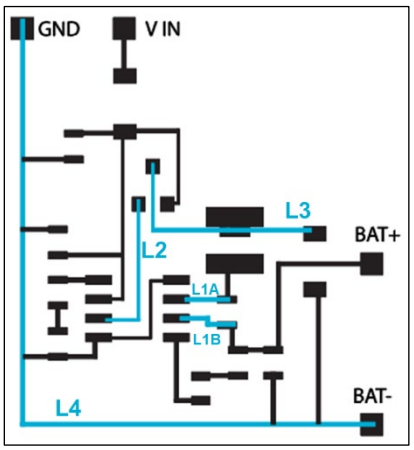

FIGURE 4: SM-PMC CHARGING CIRCUIT CAD DESIGN. HIGHLIGHTED LINES HAVE BEEN USED FOR RESISTANCE EVOLUTION MEASUREMENT

The gravure offset printing technique was used to replicate a similar sustainability study in order to investigate the feasibility of aqueous sustainable inks as compared to the traditional non-sustainable inks. For this study, the full wave rectifier circuit (similar to aerosol jet sustainability study), and a switch-mode buck battery charging circuit (switch-mode power management circuit (SM-PMC)) were developed. The full wave rectifier circuit developed using the gravure offset technique was also subjected to the repairability study and was also compared with that printed using the non-sustainable ink. The SM-PMC charging IC had an SOIC topology (pin description in TABLE 2) and the circuit designed to supply an output current of 50 mA, was developed using a sustainable ink and has been used a pure demonstrator circuit (see FIGURE 3 (C)). The developed SM-PMC circuit was then tested against a constant voltage electronic load set at 3.7V, which is the nominal voltage for a standard Li-ion battery. Assuming the Li-Ion battery capacity to be 50 mAh, the circuit was designed to charge the battery at 1C (50 mA expected output current) and was tested for a total of 50 cycles i.e. 50 hours. The parameters used for circuit evaluation were the line resistance evolution of five lines highlighted in FIGURE 4. The line resistance was measured every 10 cycles to see if the circuit current affects the same. Furthermore, the second parameter used for evaluation was the output charge current which was compared against the design value of 50 mA.

2.3 Conductive Ink Properties

The circuits for the voltage-controlled oscillator and full wave rectifier were printed using water-based sustainable silver ink A1. Ink A1 was printed using the pneumatic atomizer of the Optomec AJ300 aerosol jet system. FWRs produced using non-sustainable A2 ink were compared to those printed with A2 ink. The physical characteristics of both inks are listed in TABLE 3. The printing process variables for both inks are listed in TABLE 4.

TABLE 3: PHYSICAL PROPERTIES OF SUSTAINABLE AND NON-SUSTAINABLE INKS

Aerosol Jet Ink Physical Properties		
	Sustainable Ink A1	Non-sustainable Ink A2
Silver Content	60% by weight	55% by weight
Viscosity	150 cP	30 – 45 cP
Particle Size	400 – 700 nm	10 – 20 nm
Curing condition	20 min. @ 250 °C	10 min. @ 250 °C

TABLE 4: PRINTING PROCESS PARAMETERS FOR SUSTAINABLE AND NON-SUSTAINABLE INKS

Aerosol Jet Printing Process Parameters		
	Sustainable Ink A1	Non-sustainable Ink A2
Atomizer used	Pneumatic (PA)	
Pneumatic atomizer mass flow control (PAMFC)	1200 sccm	700 sccm
PAMFC - Exhaust mass flow control	50 - 100 sccm	30 – 50 sccm
Sheath Gas mass flow control (SMFC)	70 sccm	90 sccm
Stage speed	4 mm/s	5 mm/s
Platen Temperature	30 °C	50 °C
Ink Temperature	25 °C	
Nozzle Diameter	300 µm	
Standoff Height	2 mm	
No. of passes	4 (lines), 1 (pads)	

TABLE 5: PHYSICAL PROPERTIES OF SUSTAINABLE AND NON-SUSTAINABLE INKS (GRAVURE OFFSET)

Gravure Offset Ink Physical Properties		
	Sustainable Ink B1	Non-sustainable Ink B2
Silver Content	75% by weight	85% by weight
Viscosity	26000 cP	25000 cP

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Particle Size	4000 nm	500 nm
Curing condition	5 min. @ 250 °C	5 min. @ 225 °C

TABLE 6: PRINTING PROCESS PARAMETERS FOR ECA 'B'

Direct-Ink-Write Printing Parameters (ECA 'C')	
Nozzle Diameter	230 µm
Stage Speed	100 mm/s
Standoff Height	0.1-0.2 mm
No. of passes	1
Sintering temperature	150 °C
Sintering time	30 mins

TABLE 5 lists the sustainable and non-sustainable silver inks B1 and B2 used for the gravure offset printing process. An electrically conductive adhesive (ECA) 'C' was used for attaching components onto the printed circuits. A direct-write printer was used for dispensing the ECA onto the circuit pads. TABLE 6 lists the printing and sintering parameters used for the ECA.

2.4 Test Matrix

The test matrices used for this investigation are shown in this section. For each kind of circuit, three samples were created and subjected to testing. In order to compare the performance of the non-sustainable ink with that of the sustainable ink, full wave rectifier circuits were also produced using the non-sustainable ink.

TABLE 7: TEST MATRIX FOR SUSTAINABILITY AND REPAIRABILITY STUDY (AEROSOL JET)

	Sustainable Ink Performance and Repairability Investigation	Non-sustainable Ink Performance and Repairability Investigation
Voltage Controlled Oscillator	✓	
Full Wave Rectifier	✓	✓

TABLE 8: TEST MATRIX FOR SUSTAINABILITY AND REPAIRABILITY STUDY (GRAVURE OFFSET)

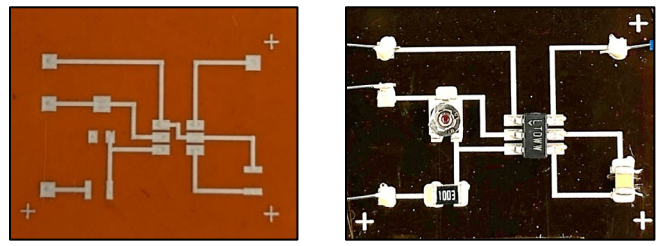
	Sustainable Ink Performance and Repairability Investigation	Non-sustainable Ink Performance and Repairability Investigation
Full Wave Rectifier	✓	✓

3. RESULTS AND DISCUSSION

The presentation of the study's findings is divided into three sections. The performance analysis and pre- and post-repairability investigation for the voltage-controlled oscillator are covered in the first section. For the full-wave rectifier, the second section has the same layout. The performance and repairability of the full wave rectifiers printed using sustainable and non-sustainable ink are compared in the last section.

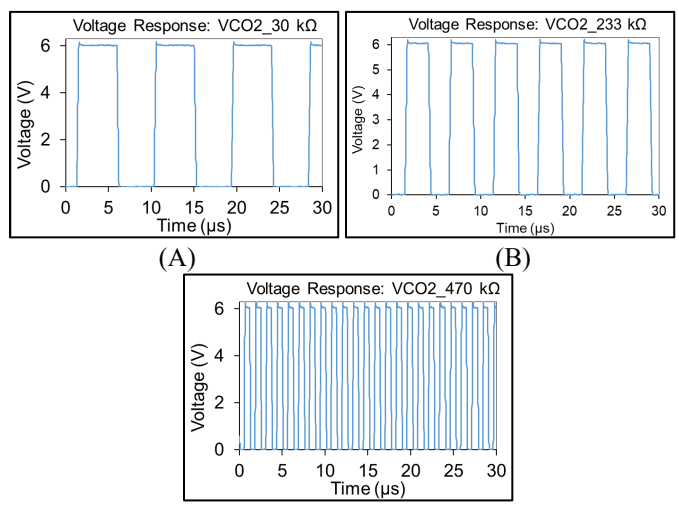
3.1 Voltage Controlled Oscillator

FIGURE 5 shows the silver conductible layer for the VCO circuit. As is the case with aerosol jet printing, the printing resolution is well-defined for the sustainable ink A1, and the IC pads do not show any interconnections with the spacing between them being distinct from one another.



(A) (B)

FIGURE 5: SILVER CONDUCTIBLE LAYER OF THE VOLTAGE CONTROLLED OSCILLATOR CIRCUIT



(A) (B) (C)

FIGURE 6: VOLTAGE-CONTROLLED OSCILLATOR OUTPUT SIGNAL FOR R_{SET} = (A) 30 KΩ, (B) 233 KΩ, (C) 470 KΩ

The voltage-controlled oscillator's output signal waveform is shown in FIGURE 6 (A), (B), and (C), respectively, for RSET values of 30 kΩ, 233 kΩ, and 470 kΩ. The frequency of the output signal increases as expected as R_{SET}'s value rises. The output signal frequency in kHz for each of the three printed voltage-controlled oscillator circuit samples is plotted against the RSET resistance value in FIGURE 7. As can be observed, the frequency response of the three samples is comparable, demonstrating the reproducible printing behavior of the sustainable silver-based ink 'A'. FIGURE 8 (A), (B), and (C) compare the pre- and post-repair performance of the three VCO samples. The oscillators' consistent performance before and after

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the repair shows that the circuits can be successfully repaired using sustainable ink.

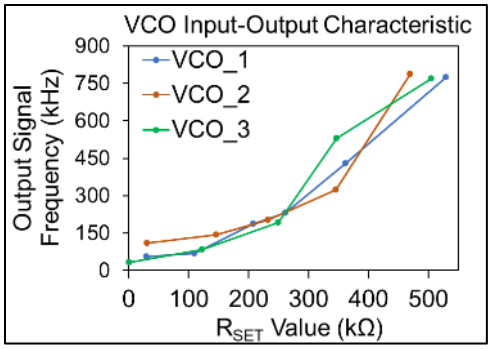
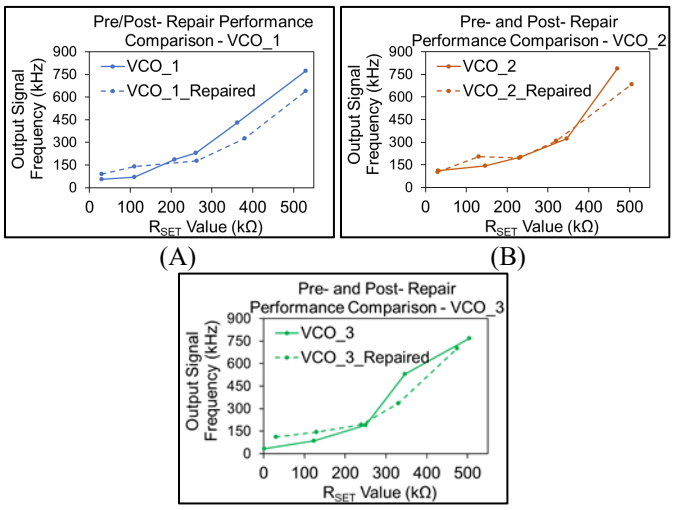


FIGURE 7: VOLTAGE CONTROLLED OSCILLATOR OUTPUT SIGNAL FREQUENCY VS. R_{SET}

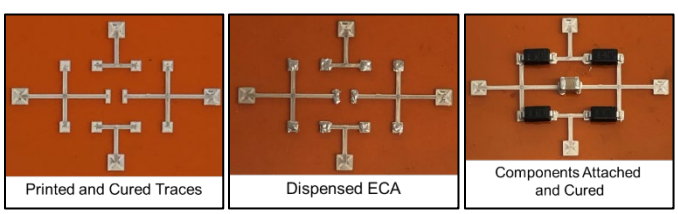


(A) (B)

(C)

FIGURE 8: PRE- AND POST-REPAIR PERFORMANCE COMPARISON FOR VCO CIRCUITS

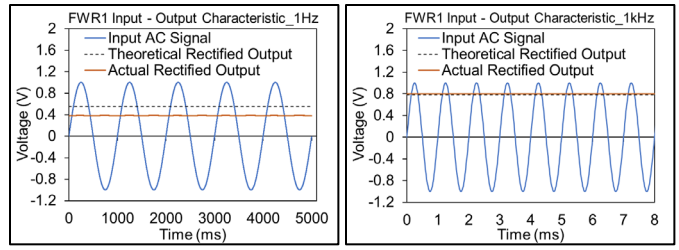
3.2 Full Wave Rectifier



(A) (B) (C)

FIGURE 9: PRISTINE CIRCUIT FABRICATION PROCESS FOR FULL WAVE RECTIFIER CIRCUIT

This section presents the performance analysis for the additively printed full-wave rectifier circuits. FIGURE 9 demonstrates the component attachment process for the full wave rectifier circuits.



(A) (B)

FIGURE 10. FULL WAVE RECTIFIER RMS OUTPUT DC VOLTAGE FOR (A) 1HZ AND (B) 1KHZ INPUT SIGNAL FREQUENCY

The RMS voltage of the rectified output signal is shown in FIGURE 10 (A) and (B) as it changes over time for various input signal frequencies. All of the graphs also include the theoretical RMS voltage of the rectified output signal for comparison. As can be observed, the actual RMS value is lower for lower input signal frequencies than the theoretical RMS value, and the theoretical RMS value is equal to or greater for input signal frequencies greater than or equal to 1 kHz. The smoothing capacitor's behavior at low frequencies is the cause of the lower RMS voltage at lower frequencies. The rectified signal from the bridge does not undergo any smoothing operation since the capacitor functions as an open circuit at low frequencies.

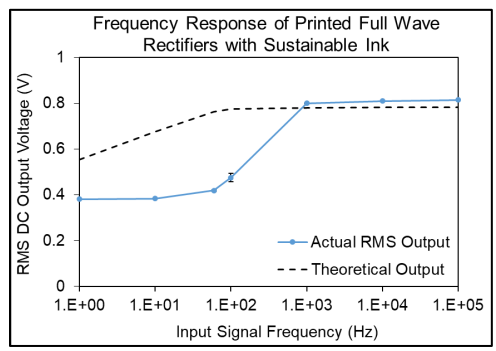
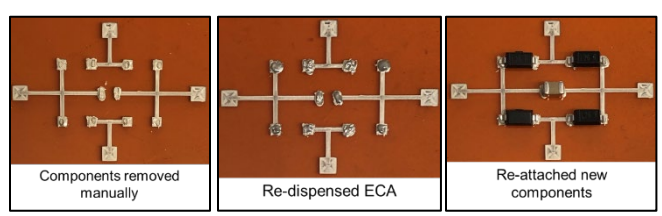


FIGURE 11: FULL WAVE RECTIFIER FREQUENCY RESPONSE (AEROSOL JET)



(A) (B) (C)

FIGURE 12: REPAIRABILITY PROCESS FOR FULL WAVE RECTIFIER CIRCUIT

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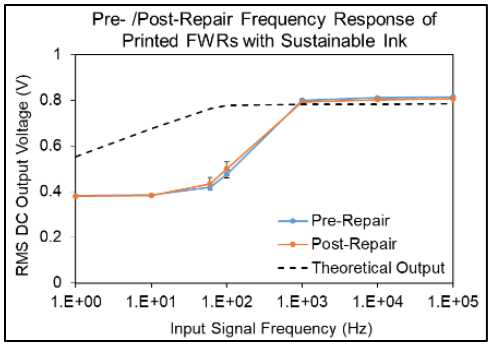

FIGURE 13: FULL WAVE RECTIFIER PRISTINE VS. POST-REPAIR FREQUENCY RESPONSE (AEROSOL JET)

The capacitor responds as expected and filters the rectified signal as the frequency of the input signal and the rectified signal grow, therefore reducing ripple and raising the total RMS value. The RMS voltage of the output DC signal is then shown against the frequency range of the input signal (1Hz to 100kHz) in FIGURE 11. For full wave rectifier circuits, a repairability analysis was also carried out. FIGURE 12 (A), (B), and (C) depict the repairability process in visual form. The frequency response of the full wave rectifier circuits before and after repair is shown in FIGURE 13. As can be seen, all three samples of full wave rectifiers exhibit strong alignment between their pre-repair and post-repair frequency response curves, demonstrating that the additively printed full wave rectifier circuit can be effectively repaired, a testament to its high sustainability.

3.3 Comparison with non-sustainable ink

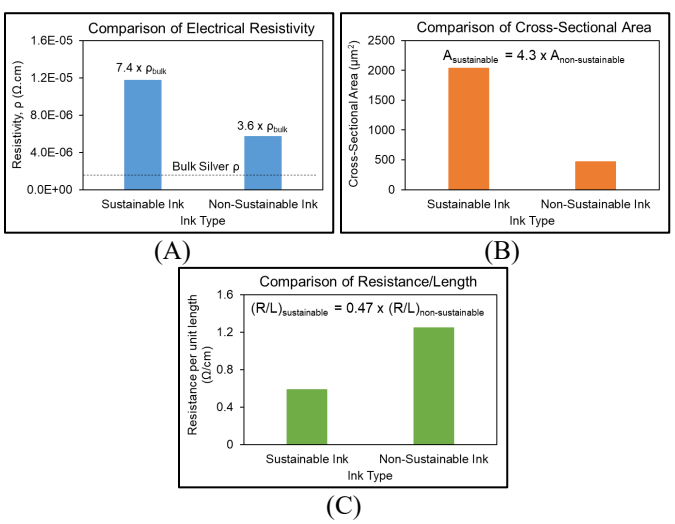

FIGURE 14: COMPARISON OF (A) ELECTRICAL RESISTIVITY, (B) CROSS-SECTIONAL AREA, AND (C) RESISTANCE PER UNIT LENGTH FOR SUSTAINABLE AND NON-SUSTAINABLE INK

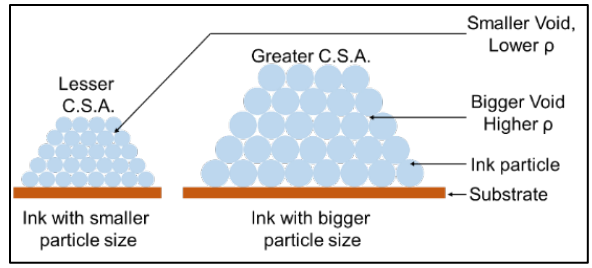

FIGURE 15: CROSS-SECTION PROFILE OF PRINTED TRACES WITH INKS HAVING SMALLER AND BIGGER PARTICLE SIZES

The full wave rectifier circuits printed with sustainable ink "A1" and those printed with non-sustainable ink "A2" are compared in this section. The traces printed with both inks also underwent a preliminary resistivity investigation. The comparison of the two inks' resistivity, cross-sectional area, and resistance per unit length is shown in FIGURE 14 (A), (B), and (C), respectively. As can be seen, the sustainable ink has approximately a quarter the cross-sectional area but half the resistivity of the non-sustainable ink. Because of this, the sustainable ink's resistance per unit length (R/L) is half that of the non-sustainable ink, making it a better option for circuit development applications.

The pre-repair performance of full wave rectifiers printed with sustainable and non-sustainable inks is compared in FIGURE 16(A). Due to the non-sustainable ink 'A2"s greater R/L value, it can be seen that the voltage output of the FWRs printed with it is slightly lower than that of the FWRs printed with sustainable ink. The pre- and post-repair performance of the FWRs produced with the non-sustainable ink is compared in FIGURE 16 (B). Contrary to the repairability analysis of the FWRs printed with the sustainable ink, which has the same pre- and post-repair performance (see FIGURE 13), the post-repair frequency response is lower than the pre-repair response. These outcomes thus serve as evidence of the usefulness and improved performance of the sustainable ink as well as the repairability of the printed circuits.

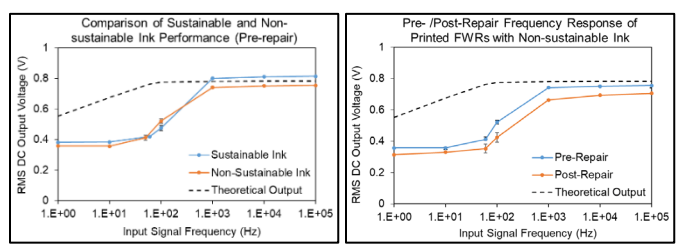

FIGURE 16: (A) PRE-REPAIR PERFORMANCE COMPARISON OF FULL WAVE RECTIFIERS PRINTED WITH SUSTAINABLE AND NON-SUSTAINABLE INKS, (B) PRE/POST-REPAIR PERFORMANCE COMPARISON WITH NON-SUSTAINABLE INK (AEROSOL JET)

3.4 Gravure Offset Circuits

FIGURE 17 depicts the pre-repair comparison of the full wave rectifiers printed by Gravure Offset method. As can be seen the frequency response curves overlap for the circuits printed

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using the sustainable and non-sustainable inks. FIGURE 18 depict the pre-repair and post-repair performance of the printed full wave rectifier circuits with the sustainable ink. There is little difference in the circuit performance post-repair similar to what was seen in the aerosol jet circuits.

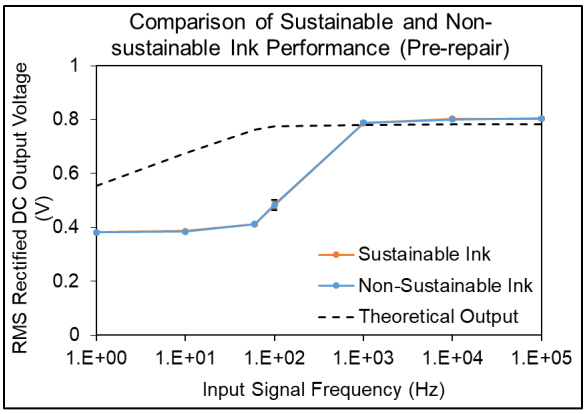

FIGURE 17: COMPARISON OF GRAVURE OFFSET PRINTED FWRs W/ SUSTAINABLE AND NON-SUSTAINABLE INK

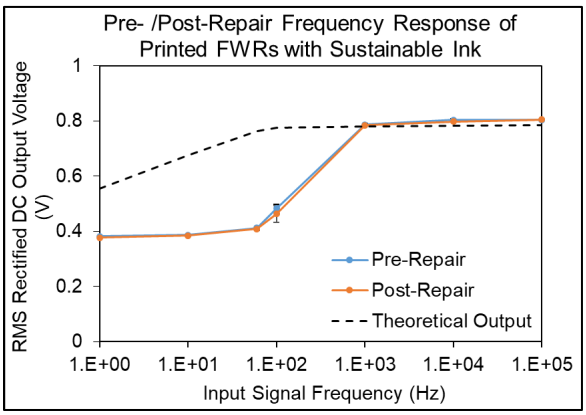

FIGURE 18: REPAIRABILITY INVESTIGATION OF GRAVURE OFFSET-PRINTED FULL WAVE RECTIFIER CIRCUITS (SUSTAINABLE INK)

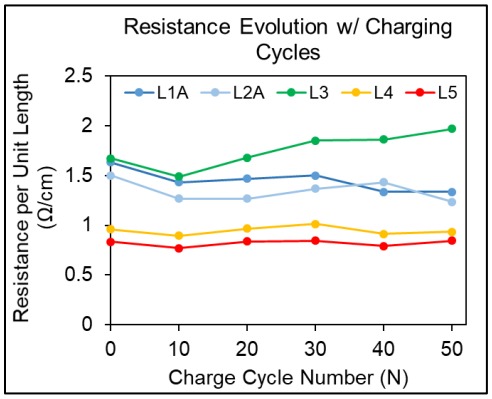

FIGURE 19: EVOLUTION OF LINE RESISTANCE PER UNIT LENGTH WITH CHARGING CYCLES

FIGURE 19 shows the line resistance evolution plots of the SM-PMC charging circuits. As can be seen, across the 50 cycles, the line resistance per unit length of the five lengths measured doesn't change significantly except for line L2. FIGURE 20 shows the evolution of the output charge current of the charging circuit. It can be seen that the output current almost matches the expected current value and is off by ~2 mA. The slight reduction in the output current occurs due to the added resistance of the printed lines and the ECA interconnect. FIGURE 21 shows the image of the printed switch mode battery charging circuit.

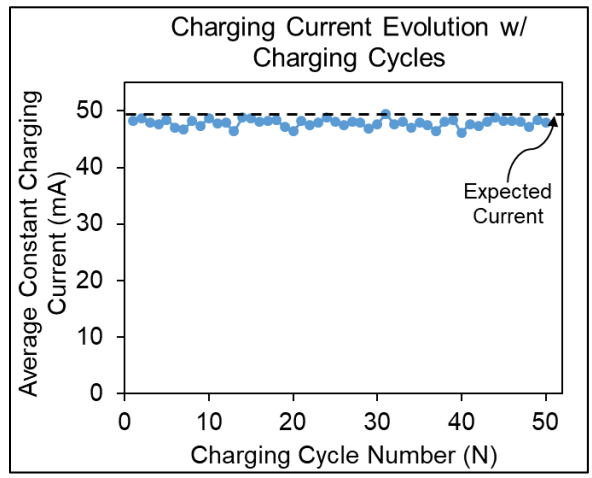

FIGURE 20: EVOLUTION OF AVERAGE CONSTANT CHARGING CURRENT WITH CHARGING CYCLES

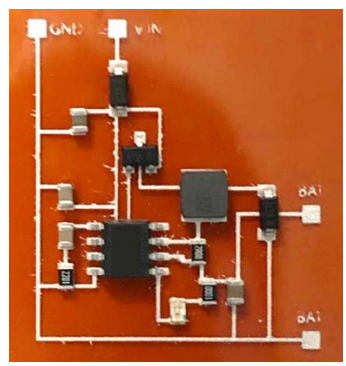

FIGURE 21: GRAVURE OFFSET PRINTED SWITCH MODE BATTERY CHARGING CIRCUIT

4. SUMMARY AND CONCLUSIONS

The durability and repairability of aerosol jet printed and gravure offset printed additive flexible circuits were successfully studied in this work. For this work, three demonstrator circuits — a full wave diode bridge rectifier, a voltage-controlled oscillator, and a switch mode battery charging circuit — were printed. For this experiment, conductible traces were printed using water-based silver ink A1 and B1, as this ink's low VOC content helps to lessen the detrimental effects of VOCs on the environment. Additionally, a repairability investigation of the VCO and FWR circuits was carried out, in which the old components were manually removed and replaced after applying ECA to the circuit

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pads. Three replicates of the input-output characteristics of the damaged and repaired circuits were found to be almost similar, demonstrating the remarkable repairability of these additive flexible circuits printed by an aerosol jet. In the end, the performance of FWR was compared to those printed with non-sustainable ink and put through a repairability investigation. In comparison to non-sustainable ink, it has been found that sustainable ink performs wither same or better in terms of both pristine performance and repairability performance. The battery charging circuit printed using the gravure offset technique also supplied the output current close to the design current value of 50 mA.

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